Challenge Problem Set 2

Math 292 Spring 2026
Rutgers University

Exercise 1:

We can show that $r(\mathbf{a}(t))$, $s(\mathbf{a}(t))$ and $q(\mathbf{a}(t))$ are all independent of t, by checking that when we take the derivative w.r.t. t, we get zero.

$$\frac{d}{dt}(r(\mathbf{a}(t))) = 2aa' + 2bb' = 2ab - 2ab = 0 \tag{1}$$
$$\frac{d}{dt}(s(\mathbf{a}(t))) = 2cc' + 2cc' = 2cd - 2cd = 0 \tag{2}$$
$$\frac{d}{dt}(q(\mathbf{a}(t))) = a'c + ac' + b'd + bd' = bc + ad - ad - bc = 0 \tag{3}$$

Exercise 2:

Let $\theta(t)$ be the angle between $\mathbf{v}(t)$ and $\mathbf{w}(t)$, we utilize the definition of the dot product

$$\begin{aligned}\cos(\theta(t)) &= \frac{\mathbf{v}(t)\cdot\mathbf{w}(t)}{||\mathbf{v}(t)||\ ||\mathbf{w}(t)||} \\ &= \frac{q}{r_0 s_0}\end{aligned}$$

In the previous exercise, we found that r, s, and q are all independent of t, thus θ must also be independent of t.

Writing the vectors in polar, we have

$$\mathbf{v}(t) = \begin{pmatrix} r_0\cos(\theta_v(t)) \\ r_0\sin(\theta_v(t))\end{pmatrix}$$
$$\mathbf{w}(t) = \begin{pmatrix} s_0\cos(\theta_w(t)) \\ s_0\sin(\theta_w(t))\end{pmatrix}$$

Differentiating with respect to t by component, we get

$$\begin{aligned}a'(t) &= -r_0\theta_v'(t)sin(\theta_v(t)) \\ b'(t) &= r_0\theta_v'(t)cos(\theta_v(t)) \\ c'(t) &= -s_0\theta_w'(t)sin(\theta_w(t)) \\ d'(t) &= s_0\theta_w'(t)cos(\theta_w(t))\end{aligned}$$

Then, substituting with $\frac{d}{dt}(a,b,c,d) = (b,-a,d,-c)$, we have

- For $a' = b$

$$\begin{aligned}-r_0\theta_v'(t)sin(\theta_v(t)) &= r_0\sin(\theta_v(t)) \\ -\theta_v'(t)sin(\theta_v(t)) &= \sin(\theta_v(t)) \\ \theta_v'(t)sin^2(\theta_v(t)) &= -\sin^2(\theta_v(t))\end{aligned}$$

- For $b' = -a$

$$r_0\theta_v'(t)cos(\theta_v(t)) = -r_0\cos(\theta_v(t))$$
$$\theta_v'(t)cos^2(\theta_v(t)) = -\cos^2(\theta_v(t))$$

Adding these first two equations, we have

$$\theta_v'(t)(sin^2(\theta_v(t)) + cos^2(\theta_v(t))) = -(\cos^2(\theta_v(t)) + \sin^2(\theta_v(t)))$$
$$\theta_v'(t) = -1$$

Similarly for $\theta_w'(t)$, we have

- For $c' = d$

$$-s_0\theta_w'(t)sin(\theta_w(t)) = s_0\sin(\theta_w(t))$$
$$\theta_w'(t)sin^2(\theta_w(t)) = -\sin^2(\theta_w(t))$$

- For $d' = -c$

$$s_0\theta_w'(t)cos(\theta_w(t)) = -s_0\cos(\theta_w(t))$$
$$\theta_w'(t)cos^2(\theta_w(t)) = -\cos^2(\theta_w(t))$$

Now, adding these two equations, we have

$$\theta_w'(t)(sin(\theta_w(t)) + cos^2(\theta_w(t))) = -(\sin^2(\theta_w(t)) + \cos^2(\theta_w(t)))$$
$$\theta_w'(t) = -1$$

Thus, we have that

$$\theta_v(t) = -t + \theta_v(0)$$
$$\theta_w(t) = -t + \theta_w(0).$$

and that

$$\begin{aligned}\theta(t) &= \theta_v(t) - \theta_w(t)\\ &= -t + \theta_v(0) - (-t + \theta_w(0))\\ &= \theta_v(0) - \theta_w(0)\end{aligned}$$

Since the scaled vectors $s_0\mathbf{v}(t)$ and $r_0\mathbf{w}(t)$ share a constant magnitude and a fixed signed angular difference $\theta = \theta_v(0) - \theta_w(0)$, the relationship between them is defined by the unique, time-independent rotation matrix. □

Exercise 3:

We show that $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$, and $\mathbf{u}_4$ form an orthonormal basis by showing that each vector is

- Unit length:

$$||\mathbf{u}_1||^2 = \frac{1}{r^2}(s_0^2 + r_0^2cos^2(\theta) + r_0^2sin^2(\theta)) = \frac{r^2}{r^2} = 1$$
$$||\mathbf{u}_2||^2 = \frac{1}{r^2}(s_0^2 + r_0^2cos^2(\theta) + r_0^2sin^2(\theta)) = \frac{r^2}{r^2} = 1$$
$$||\mathbf{u}_3||^2 = \frac{1}{r^2}(r_0^2 + s_0^2cos^2(\theta) + s_0^2sin^2(\theta)) = \frac{r^2}{r^2} = 1$$
$$||\mathbf{u}_4||^2 = \frac{1}{r^2}(r_0^2 + s_0^2cos^2(\theta) + s_0^2sin^2(\theta)) = \frac{r^2}{r^2} = 1$$

- Orthogonality:

$$\begin{aligned}
\mathbf{u}_1 \cdot \mathbf{u}_2 &= \frac{1}{r^2}(r_0^2 \cos(\theta)\sin(\theta) - r_0^2 \cos(\theta)\sin(\theta)) = \frac{0}{r^2} = 0 \\
\mathbf{u}_1 \cdot \mathbf{u}_3 &= \frac{1}{r^2}(r_0 s_0 - r_0 s_0 \cos^2(\theta) - r_0 s_0 \sin^2(\theta)) = \frac{r_0 s_0 - r_0 s_0}{r^2} = 0 \\
\mathbf{u}_1 \cdot \mathbf{u}_4 &= \frac{1}{r^2}(-r_0 s_0 \cos(\theta)\sin(\theta) + r_0 s_0 \cos(\theta)\sin(\theta)) = \frac{0}{r^2} = 0 \\
\mathbf{u}_2 \cdot \mathbf{u}_3 &= \frac{1}{r^2}(-r_0 s_0 \cos(\theta)\sin(\theta) + r_0 s_0 \cos(\theta)\sin(\theta)) = \frac{0}{r^2} = 0 \\
\mathbf{u}_2 \cdot \mathbf{u}_4 &= \frac{1}{r^2}(r_0 s_0 - r_0 s_0 \sin^2(\theta) - r_0 s_0 \cos^2(\theta)) = \frac{r_0 s_0 - r_0 s_0}{r^2} = 0 \\
\mathbf{u}_3 \cdot \mathbf{u}_4 &= \frac{1}{r^2}(s_0^2 \cos(\theta)\sin(\theta) - s_0^2 \cos(\theta)\sin(\theta)) = \frac{0}{r^2} = 0
\end{aligned}$$

Recall from the last exercise, we have the following

$$\begin{aligned}
s_0 \mathbf{v}(t) &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} r_0 \mathbf{w}_0 \\
&\implies s_0 a - r_0(c\cos\theta - d\sin\theta) = 0 \qquad (1) \\
&\implies s_0 b - r_0(c\sin\theta + d\cos\theta) = 0 \qquad (2)
\end{aligned}$$

We use this to show that $\mathbf{a}(t) \cdot \mathbf{u}_1 = \mathbf{a}(t) \cdot \mathbf{u}_2 = 0$

$$\begin{aligned}
\mathbf{a}(t) \cdot \mathbf{u}_1 &= \frac{1}{r}(a s_0 - c r_0 \cos(\theta) + d r_0 \sin(\theta)) = 0 \\
\mathbf{a}(t) \cdot \mathbf{u}_2 &= \frac{1}{r}(b s_0 - c r_0 \sin(\theta) - d r_0 \cos(\theta)) = 0
\end{aligned}$$

Since we have that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ forms an orthonormal basis in $\mathbb{R}^4$, we can write $\mathbf{a}(t)$ as a linear combination of the basis vectors with constants $c_1, c_2, c_3, c_4 \in \mathbb{R}$

$$\mathbf{a}(t) = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + c_4\mathbf{u}_4$$

We already saw that $c_1 = c_2 = 0$. Thus, replacing $\alpha(t) = c_3 = \mathbf{a}(t) \cdot \mathbf{u}_3$ and $\beta(t) = c_4 = \mathbf{a}(t) \cdot \mathbf{u}_4$, we are left with

$$\mathbf{a}(t) = \alpha(t)\mathbf{u}_3 + \beta(t)\mathbf{u}_4.$$

Then, taking the squared euclidean norm, we have that

$$||\mathbf{a}(t)||^2 = \alpha^2(t) + \beta^2(t).$$

Recall the original definition of $\mathbf{a}(t) = (a, b, c, d)$, taking the squared euclidean norm, we have that

$$\begin{aligned}
||\mathbf{a}(t)||^2 &= a^2 + b^2 + c^2 + d^2 \\
&= r_0^2 + s_0^2 \\
&= r^2
\end{aligned}$$

Thus, we have that

$$\alpha^2(t) + \beta^2(t) = r^2.$$

Because $\mathbf{a}(t)$, $\mathbf{u}_3$, and $\mathbf{u}_4$ are continuously differentiable, their dot products $\alpha(t)$ and $\beta(t)$ are also continuously differentiable.

Taking the derivative w.r.t. t, we have

- For $\alpha(t) = \mathbf{a}(t) \cdot \mathbf{u}_3$:

$$\alpha(t) = \frac{1}{r}(ar_0 + cs_0\cos(\theta) - ds_0\sin(\theta))$$

$$\begin{aligned}\alpha'(t) &= \frac{1}{r}(a'r_0 + c's_0\cos(\theta) - d's_0\sin(\theta)) \\ &= \frac{1}{r}(br_0 + ds_0\cos(\theta) + cs_0\sin(\theta)) \\ &= \beta(t)\end{aligned}$$

- For $\beta(t) = \mathbf{a}(t) \cdot \mathbf{u}_4$:

$$\beta(t) = \frac{1}{r}(br_0 + cs_0\sin(\theta) + ds_0\cos(\theta))$$

$$\begin{aligned}\beta'(t) &= \frac{1}{r}(b'r_0 + c's_0\sin(\theta) + d's_0\cos(\theta)) \\ &= \frac{1}{r}(-ar_0 + ds_0\sin(\theta) + -cs_0\cos(\theta)) \\ &= -\alpha(t)\end{aligned}$$

Exercise 4:

Let $\mathbf{a}(0) = (1, 0, 1, \sqrt{3})$. Then, we can solve for r, s, and q since they are time independent

$$\begin{aligned}r_0 &= \sqrt{a^2(0) + b^2(0)} = 1 \\ s_0 &= \sqrt{c^2(0) + d^2(0)} = 2 \\ r &= \sqrt{r_0^2 + s_0^2} = \sqrt{5}\end{aligned}$$

We use the rotation transformation from exercise 2 to solve for $\cos\theta$ and $\sin\theta$

$$\begin{aligned}s_0 a &= r_0(c\cos\theta - d\sin\theta) \\ &= 2 - \cos\theta + \sqrt{3}\sin\theta = 0 \\ s_0 b &= r_0(c\sin\theta + d\cos\theta) \qquad (1) \\ &= \sin\theta + \sqrt{3}\cos\theta = 0 \\ &\implies \sin\theta = -\sqrt{3}\cos\theta \qquad (2)\end{aligned}$$

Plugging equation 2 into 1, we have

$$2 - \cos\theta + \sqrt{3}(-\sqrt{3}\cos\theta) = 2 - 4\cos\theta \implies \cos\theta = \frac{1}{2}$$

Then, substituting back into equation 2 we have $\sin\theta = -\frac{\sqrt{3}}{2}$.
With this, we can write $\mathbf{u}_3$ and $\mathbf{u}_4$ with our initial conditions

$$\mathbf{u}_3 = \frac{1}{\sqrt{5}}\begin{bmatrix}1\\0\\1\\\sqrt{3}\end{bmatrix} = \frac{1}{\sqrt{5}}\mathbf{a}(0)$$

$$\mathbf{u}_4 = \frac{1}{\sqrt{5}}\begin{bmatrix}0\\1\\-\sqrt{3}\\1\end{bmatrix}$$

We can solve for the scalars initial state as well

$$\alpha(0) = \mathbf{a}(0)\cdot\frac{1}{\sqrt{5}}\mathbf{a}(0) = \frac{1}{\sqrt{5}}(5) = r$$

$$\beta(0) = \mathbf{a}(0)\cdot\frac{1}{\sqrt{5}}\begin{bmatrix}0\\1\\-\sqrt{3}\\1\end{bmatrix} = 0$$

Thus, we choose

$$\alpha(t) = \sqrt{5}\cos(t)$$
$$\beta(t) = -\sqrt{5}\sin(t)$$

Combining everything, we have

$$\mathbf{a}(t) = \begin{bmatrix}\cos(t)\\-\sin(t)\\\cos(t)+\sqrt{3}\sin(t)\\\sqrt{3}\cos(t)-\sin(t)\end{bmatrix}$$

Parameterizing the curve, we have

$$(x(t), y(t)) = (\cos(t), \cos(t)+\sqrt{3}\sin(t))$$

Substituting x(t) for cos(t) in y(t), then squaring, we have

$$y = x + \sqrt{3}\sin(t)$$
$$\sin(t) = \frac{y-x}{\sqrt{3}}$$
$$\sin^2(t) = \frac{y^2-2xy+x^2}{3}$$

Then, squaring x(t), we get $x^2(t) = \cos^2(t)$. Now, we add the two equations and get

$$\frac{y^2-2xy+x^2}{3} + x^2(t) = \cos^2(t)+\sin^2(t)$$
$$\frac{y^2-2xy+4x^2}{3} = 1$$
$$y^2-2xy+4x^2 = 3$$

The equation $y^2 - 2xy + 4x^2 = 3$ is exactly that of the ellipse we got in the first workshop.

Exercise 5:

Checking the derivative of $p(\mathbf{a}(t))$ w.r.t. t

$$\frac{d}{dt}(p(\mathbf{a}(t))) = a'd + ad' - b'c - bc' = bd - ac + ac - bd = 0$$

Indeed the derivative is zero, thus $p(\mathbf{a}(t))$ is independent of t.

Next, we have that $\mathbf{a}(t) \in \mathbb{S}^3$, which means $a^2 + b^2 + c^2 + d^2 = 1 \implies r + s = 1$.
We want to compute the left hand side, so we start with $p^2 + q^2$

$$\begin{aligned} p(\mathbf{a}(t))^2 + q(\mathbf{a}(t))^2 &= a^2d^2 - 2abcd + b^2c^2 + a^2b^2 + 2abcd + c^2d^2 \\ &= a^2d^2 + b^2c^2 + a^2c^2 + b^2d^2 \\ &= a^2(c^2 + d^2) + b^2(c^2 + d^2) \\ &= (a^2 + b^2)(c^2 + d^2) \\ &= r(\mathbf{a}(t)) * s(\mathbf{a}(t)) \end{aligned}$$

Then, the left hand side simplifies to

$$\begin{aligned} (2p)^2 + (2q)^2 + (r - s)^2 &= 4p^2 + 4q^2 + r^2 - 2rs + s^2 \\ &= r^2 + 2rs + s^2 \\ &= (r + s)^2 \\ &= 1^2 \\ &= 1 \end{aligned}$$

This is exactly the identity we wanted.